Report on the Project
LazyFCA Classification of Telco Customer Churn

Prepared by: Gorelskii Roman, Radaeva Karina

GitHub Link: `https://github.com/RomanGorelsky/lazy_fca_osda`

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1 Telco Dataset

1.1 Dataset Overview

The dataset used in this study is sourced from Kaggle: https://www.kaggle.com/datasets/blastchar/telco-customer-churn.

It consists of 7032 objects and 21 columns. A sample of the data is shown below:

customerID	gender	Senior Citizen	Partner	Dependents	tenure	...	Churn
7590-VHVEG	Female	0	Yes	No	1	...	No
55575-GNVDE	Male	0	No	No	34	...	No
3668-QPYBK	Male	0	No	No	2	...	Yes
7795-CFOCW	Male	0	No	No	45	...	No
1452-KIOVK	Male	0	No	Yes	22	...	No

Table 1.1: Sample of the Telco Customer Churn Dataset

Distribution of features before any preprocessing:

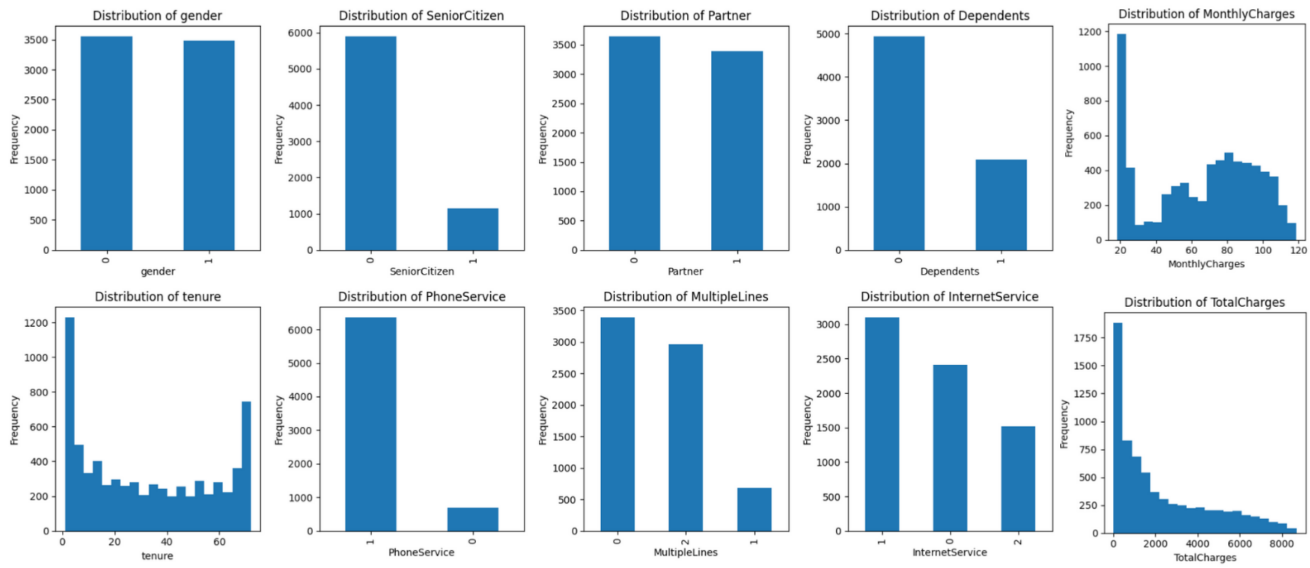

Figure 1.1: Distribution of features part 1

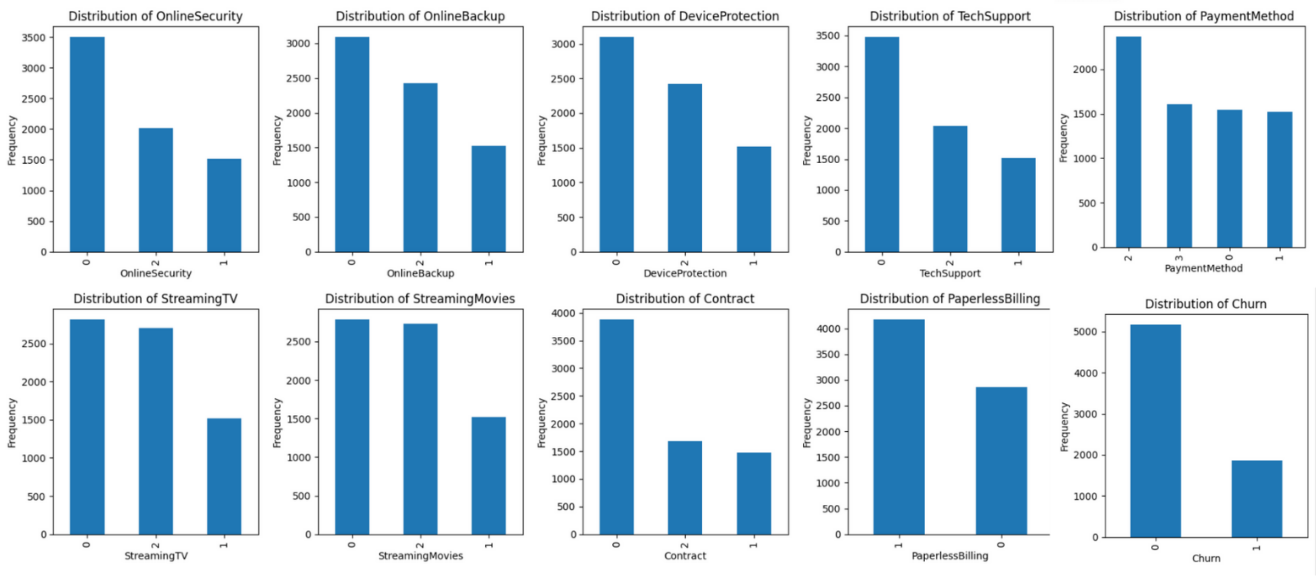

Figure 1.2: Distribution of features part 2

1.2 Dataset Preprocessing

Due to class imbalance in the target variable 'Churn', Random Undersampling was applied to balance the dataset. The final sample contains 3738 objects.

For classic models:

- Categoric features were encoded with One-Hot encoder

- Numeric features were scaled only for KNN, SVM and Logistic Regression.

For LazyFCA:

- Categoric features were binarized

- Numeric features were transformed by interval Pattern Structures.

2 Pattern Structures

The interval pattern structures were used in the model architecture. Considering the min or max versions of them, they were considered for implementation. But there was no logical explanation for their use for the existing numerical features. The interval structure suits better in the given problem statement.

3 Architecture Improvements

- **GPU acceleration:**

  2.5 hours on CPU → 50 minutes on GPU

- **Random sampling: 50%**

  50 minutes → 25 minutes on GPU

- **Redefinition of the classification rule:**

```
if negative_classifiers / (negative_classifiers + positive_classifiers) < alpha and positive_classifiers >= beta:
    return 1  # positive
elif negative_classifiers / (negative_classifiers + positive_classifiers) > alpha and negative_classifiers >= beta:
    return 0  # negative
else:
    return 1
```

α and β were tuned via GridSearch

4 Results

The following results were obtained (parameters for standard models were tuned using GridSearchCV):

	TP	TN	FP	FN	TNR	NPV	FPR	FDR
KNN	275	269	105	99	0.72	0.73	0.28	0.28
NaiveBayes	326	212	162	48	0.57	0.81	0.43	0.33
LogReg	280	277	97	94	0.74	0.75	0.26	0.26
SVM	288	269	105	86	0.72	0.76	0.28	0.27
DecisionTree	326	216	158	48	0.58	0.82	0.42	0.33
RandomForest	292	268	106	82	0.72	0.77	0.28	0.27
XGBoost	290	270	104	84	0.72	0.76	0.28	0.26
LazyFCA	277	266	108	97	0.71	0.73	0.29	0.28
LazyFCA Improved	321	212	162	53	0.57	0.80	0.43	0.34

	Accuracy	Precision	Recall	F1
KNN	0.73	0.72	0.74	0.73
NaiveBayes	0.72	0.67	0.87	0.76
LogReg	0.75	0.74	0.75	0.75
SVM	0.75	0.73	0.77	0.75
DecisionTree	0.73	0.67	0.87	0.76
RandomForest	0.75	0.73	0.78	0.76
XGBoost	0.75	0.74	0.78	0.76
LazyFCA	0.73	0.72	**0.74**	**0.73**
LazyFCA Improved	0.71	0.67	**0.86**	**0.75**

Table 4.1: Performance of Improved LazyFCA Model

5 Interpretability analysis

5.1 Decision Tree Interpretability

Some of the standard models tend to be highly interpretable. For example, Decision Tree is a white box that can be clearly visualized (represented on the Figure (5.1)).

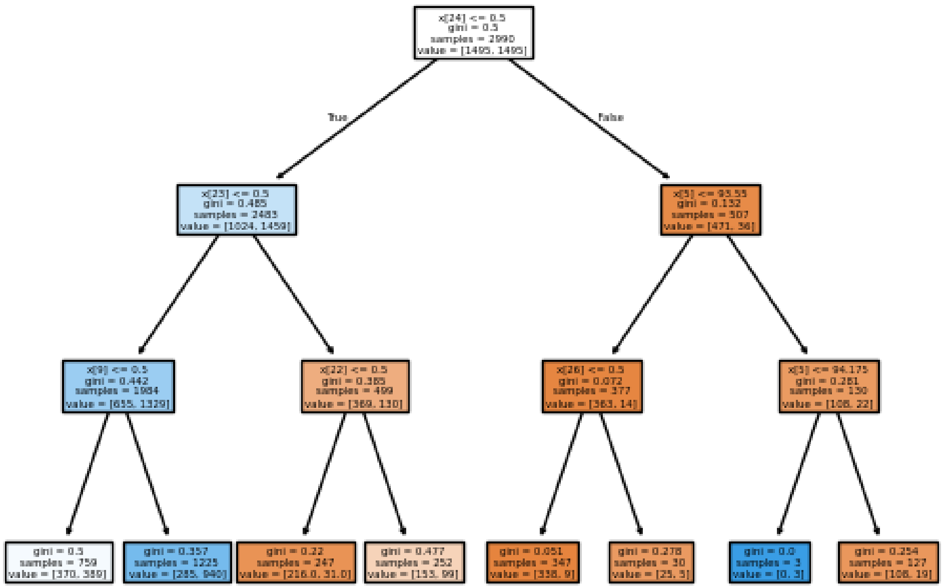

Figure 5.1: Decision Tree Plot

Thus, there is a need to extract the reasons for classification of a sample from our that would allow to evaluate its explanability as well.

5.2 Averaged Explanations over Entire Dataset

Some statistics over the entire dataset:

- Average proportion of positive matches: 52.9%
- Average proportion of negative matches: 47.1%
- Average confidence: 75.06%

From this statistics it can be seen that the improvements of redefining the classification rule help to overcome the rest ∼25% of the confidence from the classification of nearly 50:50 cases matching.

5.3 Explanations for Specific Samples

The local explainability can be derived for each particular sample of the test dataset.

Example: Explanations for Sample 97. This sample is classified as positive with 75.3% confidence. It has 140 supporting classifiers and 46 opposing classifiers.

- Key binary feature: `'fiber_optic'` appears in 69% of supporters.
- Strongest supporter has 18 matching samples.
- To flip the prediction, the most impactful change would be: Flip `no_online_security` from True to False.

5.4 Most Influential Features

Another global explainability feature (common for many models) is the feature importance extraction (Figure (5.2)).

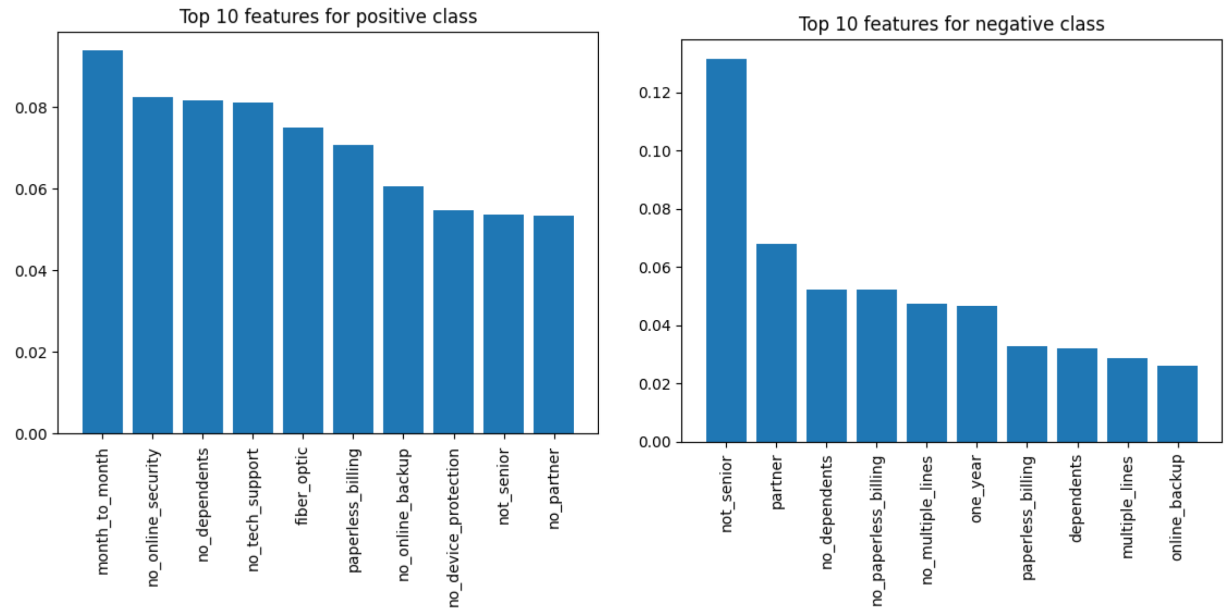

Figure 5.2: Most influential features for positive an negative classes from LazyFCA

6 Conclusion

The introduced modifications help increase the speed of the classification procedure and enhance the F1-score by placing greater emphasis on Recall.

In addition, the model provides both global explanations and local explanations for specific samples, revealing the underlying classification rules.

7 Future Improvements

Future studies could focus on introducing new modifications to improve Accuracy and Precision, making the architecture more robust across various classification datasets.